

THE SAME EIGHT-CELL DESIGN—WITH UNEQUAL EVIDENCE

Each model crosses transition type × comparison design × temporal scope

DEEPSEEK

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

Local = window around the first divergence—not a semantic error category.

CODELLAMA

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ONLY 4 POSITIVE TRANSITIONS

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

CodeLlama improvement cells were excluded from candidate selection.

A SCREEN PRODUCES A RANKED, AUDITABLE SHORTLIST

DeepSeek · improvement association · global · base-enriched

Rank	Feature	Summary	Effect	E/V	Support
1	3602	active_fraction	-0.021	-8.23	187
2	10168	active_fraction	-0.018	-7.67	165
3	5422	active_fraction	-0.005	-7.66	139
4	1982	active_fraction	-0.015	-7.53	200
5	1862	active_fraction	-0.008	-7.45	168
6	1650	active_fraction	-0.026	-7.28	180
7	5039	mean	-0.088	-7.26	147
8	9580	active_fraction	-0.013	-7.24	191
9	5837	active_fraction	-0.010	-7.18	166
10	9537	active_fraction	-0.038	-7.15	158

E/V = observed contrast ÷ SD of the label-permutation null

Signal-to-noise score—not a z-score, p-value, corrected significance test, or causal estimate.

THE SAME OUTPUT FORMAT APPLIES TO CODELLAMA

CodeLlama · regression association · global · absolute ranking

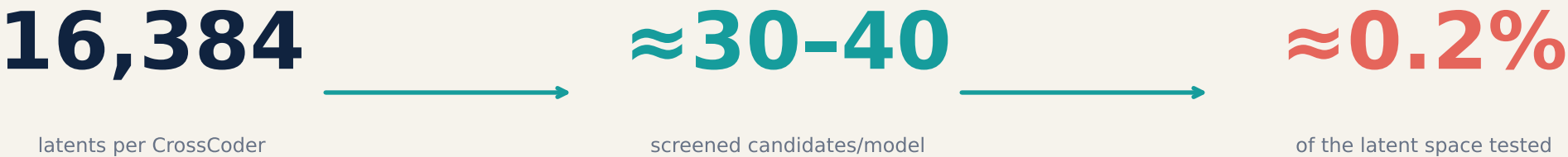
Rank	Feature	Summary	Effect	E/V	Support
1	13147	mean	+0.126	+5.24	95
2	8992	active_fraction	-0.011	-4.93	136
3	1840	active_fraction	-0.046	-4.77	291
4	16161	mean	+0.127	+4.76	33
5	15471	active_fraction	+0.005	+4.75	55
6	10570	mean	+0.188	+4.60	53
7	8138	max	-4.973	-4.51	291
8	7353	mean	+0.228	+4.48	54
9	14359	mean	-0.166	-4.46	286
10	2825	mean	+0.338	+4.38	92

E/V = observed contrast ÷ SD of the label-permutation null

Signal-to-noise score—not a z-score, p-value, corrected significance test, or causal estimate.

SCREEN FIRST; INTERVENE AT A COMMON OPERATING POINT

The associative stage makes causal testing computationally tractable



WHY $|\alpha| = 3$?

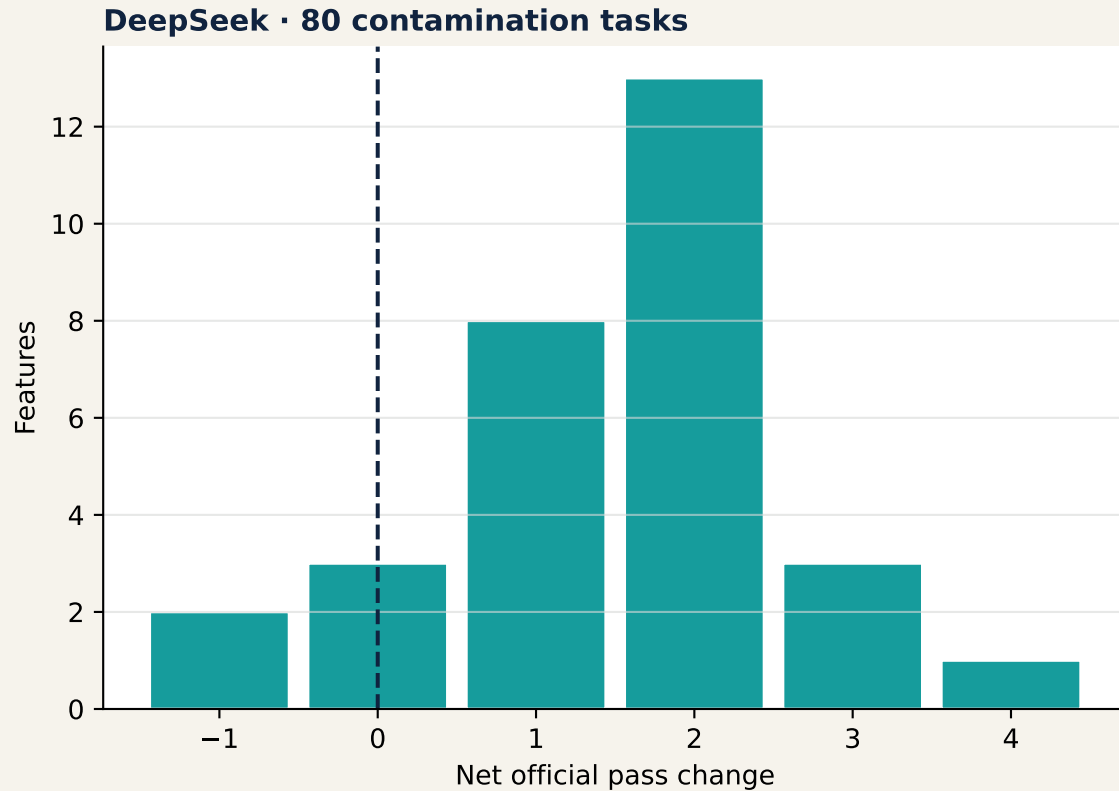
Earlier dose-response experiments suggested a practical exploratory point: visible trajectory changes below the standardized disruptive high-dose region, dozens of candidates before full dose curves, reverse-mode

WHY ONE COMMON MAGNITUDE?

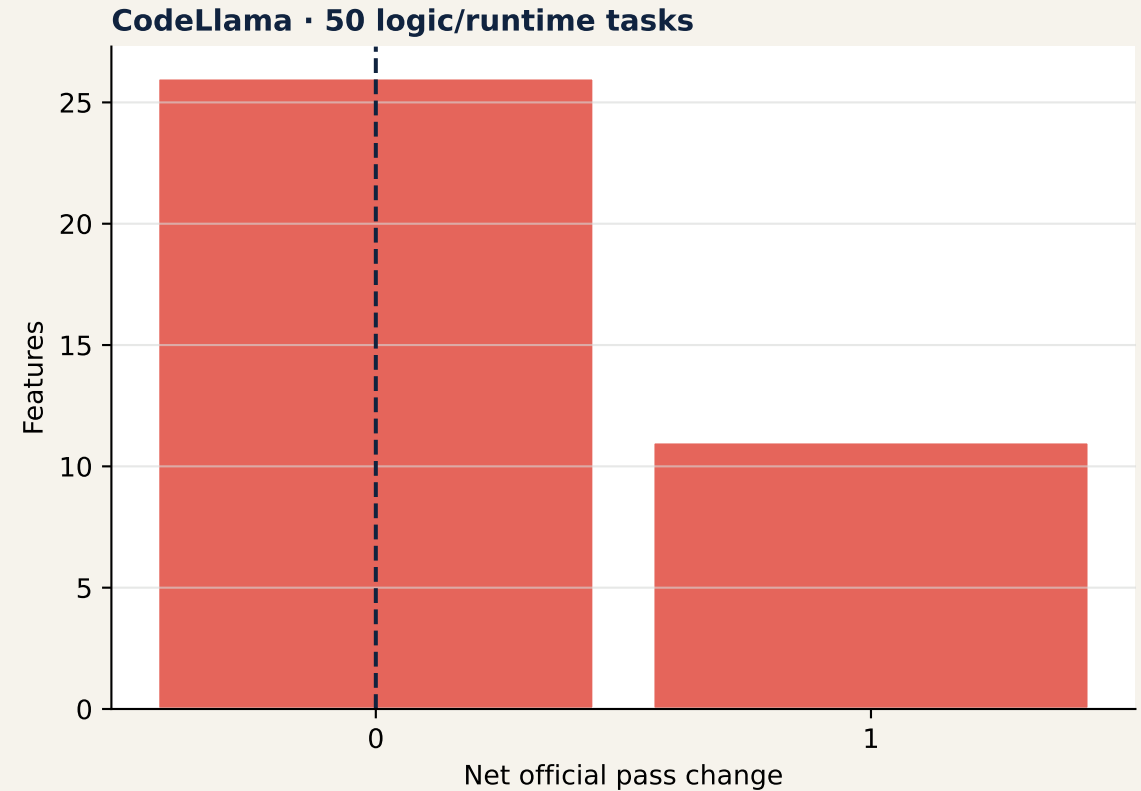
$|\alpha|=3$ is an exploratory operating point—not an independently validated optimum.

THE $\alpha=3$ RESPONSE DISTRIBUTIONS LOOK VERY DIFFERENT

Official net pass change relative to each cohort baseline



DeepSeek: best +4; many positive arms



CodeLlama: most arms 0; best +1

Consistent with more heterogeneous/distributed tested CodeLlama failures—not proof of greater network-wide damage.

DID WE FIND CAUSAL FEATURES—OR STEERABLE TASKS?

Success concentration makes placebo and alternative-latent controls essential

DEEPSEEK /1030

Corrected by 29 / 30 tested features.
A broadly sensitive task can inflate many candidate directions.

CODELLAMA /119

Responsible for 10 / 11 positive arms.
A fail→pass transition alone cannot establish specificity.

TARGET FEATURE

Behavior-aligned latent selected by screening.

RANDOM LATENTS

Not selected using screening or prior $\alpha=3$ outcomes.

ORTHOGONAL SHAMS

Random directions with controlled norm and geometry.

BBASV tests superiority to generic perturbations and coherent reversal across models.

TEN CANDIDATES ADVANCE TO BBASV

Official $\alpha=3$ outcome first; prior applicable screening rank breaks causal ties

Model	Feature	Best prior screen	Summary	E/V	$\alpha=3$	Net Δ	BBASV direction
DeepSeek	3048	paired global	active frac.	−9.97	19/80	+4	base − / FT +
DeepSeek	13801	paired global	max	−10.73	18/80	+3	base − / FT +
DeepSeek	7828	association global	mean	−7.01	18/80	+3	base − / FT +
DeepSeek	15669	paired global	max	−10.06	18/80	+3	base − / FT +
DeepSeek	13191	paired global	max	−11.03	17/80	+2	base − / FT +
CodeLlama	13147	association global	mean	+5.24	1/50	+1	merged − / base +
CodeLlama	12253	paired global	max	−12.73	1/50	+1	merged + / base −
CodeLlama	10570	association global	mean	+4.60	1/50	+1	merged − / base +
CodeLlama	14359	association global	mean	−4.46	1/50	+1	merged + / base −
CodeLlama	2310	association global	active frac.	−4.30	1/50	+1	merged + / base −

Behavior-aligned direction in the worse model + reversed direction in the better model + random latents + orthogonal shams.

Screen support belongs to the broad transition population; causal outcomes use narrower semantic cohorts.